Automated Sleep Staging of a Rodent EEG/EMG Dataset Using AccuSleePy: A Complete Analysis Pipeline

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March 2026

Abstract

This report describes the application of AccuSleePy, a Python-based automated rodent sleep staging tool, to a publicly available mouse EEG/EMG dataset [Barger et al., 2019, Barger and Frye, 2025] (https://osf.io/py5eb/). The dataset comprises 50 four-hour daytime recordings from 10 C57BL/6 mice (5 recordings per animal), sampled at 512 Hz with expert manual sleep-stage labels provided as ground truth. Sleep staging was performed using AccuSleePy's compact neural network model for 2.5-second epochs (`ssann_2(5)s.pth`). Distributional shift correction was applied to each recording using a calibration set of 120 epochs per sleep stage (360 epochs total), selected using evenly distributed sampling across each stage's occurrence indices in the expert labels. Validation on held-out epochs (those not used in calibration) yielded a mean Cohen's kappa of 0.9490 ± 0.0148 and a mean accuracy of $97.25 \pm 0.72\%$, consistent with the benchmark reported in the original AccuSleePy publication [Barger et al., 2019]. No recordings were flagged by quality control. Group-level sleep architecture showed a mean of $34.53 \pm 7.76\%$ Wake, $54.64 \pm 6.16\%$ NREM, and $10.83 \pm 2.18\%$ REM across all 50 recordings, with mean bout durations of 41.5 ± 15.9 s (Wake), 62.9 ± 9.9 s (NREM), and 76.4 ± 16.6 s (REM). The complete analysis package, including scored label files, per-recording metrics, and publication-quality figures, is provided alongside this report.

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1 Data

1.1 Dataset

The dataset is drawn from the Open Science Framework repository of Barger and Frye [Barger and Frye, 2025] and is associated with the publication by Barger et al. [Barger et al., 2019]. The dataset consists of standard cortical EEG and nuchal EMG recordings from 10 adult male C57BL/6 mice, each recorded across five separate four-hour daytime sessions (50 recordings total). EEG and EMG signals were sampled at 512 Hz. Expert manual sleep-stage labels are provided at a resolution of 2.5 seconds per epoch (5,760 epochs per recording), classified into three states: REM sleep (label 1), Wake (label 2), and NREM sleep (label 3).

1.2 Recording Specifications

Table 1: Dataset summary.

Property	Value
Species	C57BL/6 mouse
Number of animals	10
Recordings per animal	5 (daytime, 4-hour)
Total recordings	50
Sampling rate	512 Hz
Epoch length	2.5 s (1,280 samples/epoch)
Epochs per recording	5,760
Total epochs	288,000
EEG configuration	Standard cortical EEG
EMG configuration	Nuchal (neck) EMG
Label encoding	REM = 1; Wake = 2; NREM = 3

1.3 Label Distribution

Table 2: Expert manual label distribution across all 50 recordings (288,000 epochs total).

Label	Stage	Count	Percentage
1	REM	30,128	10.46%
2	Wake	98,983	34.37%
3	NREM	158,889	55.17%
	Total	**288,000**	**100.00%**

2 Methods

2.1 Data Loading

All EEG and EMG signals are stored in Apache Parquet format (`recording.parquet`; shape 7,372,800 × 2, dtype float64) with columns `eeg` and `emg`. Expert manual labels are

stored in CSV format (`labels.csv`; column `brain_state`, dtype int64; 5,760 rows per recording). All loading operations are handled by `scripts/utils/data_loading.py`, a shared utility module used by every script in the pipeline. This module wraps `accusleepy.fileio.load_recording` (for EEG/EMG signals) and `accusleepy.fileio.load_labels` (for label arrays), returning standard 1-D float64 and int64 NumPy arrays respectively. Centralising all I/O in this module ensures consistent loading behaviour across all pipeline stages.

2.2 Calibration

AccuSleePy includes a distributional shift correction step (calibration) that uses a small set of manually labeled epochs from the target recording to adjust the model's decision boundaries before scoring [Barger et al., 2019]. This step is a core part of the standard AccuSleePy workflow and was applied to every recording in this analysis.

For each recording, a calibration set of 360 epochs was constructed as follows. For each of the three sleep stages (Wake, NREM, REM), all epoch indices carrying that stage label in the expert labels were identified. A subset of 120 epochs was then selected using evenly distributed positions:

```
np.linspace(0, len(stage_indices) - 1, 120, dtype=int)
```

indexed into the array of per-stage occurrence indices. This distributes the selected calibration epochs evenly across the full temporal span of each stage's occurrences, capturing within-session variability in neural signal statistics. The three per-stage sets (120 epochs each) were combined to form the final 360-epoch calibration set. The AccuSleePy built-in calibration function was applied using these epochs and their corresponding expert labels before any scoring took place. Calibration epoch indices were saved as companion files alongside the predicted-label outputs and were excluded from all subsequent validation comparisons.

2.3 Scoring

All epochs in each recording were scored using AccuSleePy's pre-trained 2.5-second model (`ssann_2(5)s.pth`) applied after calibration. AccuSleePy produces one predicted label and one confidence score per epoch. Output files were saved in CSV format (columns `brain_state` and `confidence_score`; 5,760 rows per recording) to the `outputs/predicted_labels/` directory. No intermediate disk files are created by AccuSleePy during scoring; all computations are performed in memory.

2.4 Quality Control

Quality control was applied to each recording's predicted label sequence using the following criteria:

- **Stage proportion plausibility:** a recording was flagged if Wake exceeded 80%, REM exceeded 25%, or NREM fell below 10%.
- **Long unbroken runs:** a recording was flagged if any single stage persisted without interruption for more than 60 minutes.

- **Low-confidence epochs:** epochs with a confidence score ≤ 0.8 were catalogued. For each recording, a CSV file listing all such epochs (columns: `epoch_index`, `predicted_label`, `confidence_score`) was written to the `low_confidence_epochs/` directory.

Flagged recordings were noted in `QC_report.md` but were not excluded from any analysis. The decision to exclude or retain flagged recordings is left to the researcher.

2.5 Validation Against Expert Labels

For each recording, the predicted label sequence was compared to the expert manual labels on the set of held-out epochs (all epochs not included in the calibration set; 5,400 per recording). The following metrics were computed using shared utilities in `scripts/utils/metrics.py`:

- **Cohen's kappa** — the primary validation metric; accounts for class imbalance.
- **Overall accuracy** — fraction of correctly classified held-out epochs.
- **Per-class precision, recall, and F1** for each of Wake, NREM, and REM.
- **Confusion matrix** — 3×3 counts (rows = expert label; columns = predicted label) for each recording.

Aggregate statistics (mean $\pm$ SD across all 50 recordings) were computed for kappa, accuracy, and per-class F1.

2.6 Descriptive Sleep Metrics

The following metrics were computed per recording from the predicted label sequence (not from expert labels):

- **Stage proportions:** percentage of total epochs classified as Wake, NREM, and REM.
- **Bout analysis:** a bout is defined as a continuous, uninterrupted run of epochs assigned to the same stage. For each stage, mean bout duration, maximum bout duration, and total bout count were computed. Durations are expressed in seconds (epoch length = 2.5 s).
- **State transition probability matrix:** for each pair of stages (i, j), the proportion of all transitions out of stage i that lead to stage j. Rows are normalized to sum to 1.
- **Low-confidence epoch summary:** per-recording count and percentage of epochs with confidence score ≤ 0.8.

All metrics were saved to `outputs/sleep_metrics.csv` (50 rows $\times$ 26 columns).

2.7 Software

Table 3: Software environment.

Component	Version
Python	3.11.9
AccuSleePy	0.12.0
NumPy	2.4.3
SciPy	1.17.1
pandas	2.3.3
matplotlib	3.10.8
PyTorch	2.10.0
torchvision	0.25.0

Full pinned dependency versions are available in `requirements.txt`.

3 Results

3.1 Pipeline Validation

Validation is presented as a technical quality check confirming that the AccuSleePy pipeline executed correctly on this dataset. It is not the primary scientific deliverable of this analysis.

Predicted labels were compared to expert manual labels on held-out epochs for all 50 recordings. The aggregate 3×3 confusion matrix (summed across all recordings, normalized to row percentages) is shown in Figure 1. The distribution of per-recording Cohen's kappa values is shown in Figure 2.

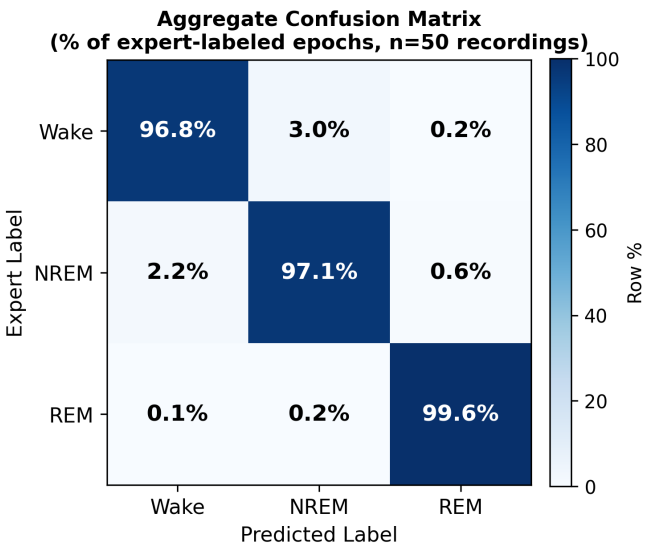

Figure 1: Aggregate confusion matrix across all 50 recordings (row-normalized to percentages). Rows represent expert labels; columns represent AccuSleePy predicted labels. Stage order: Wake, NREM, REM.

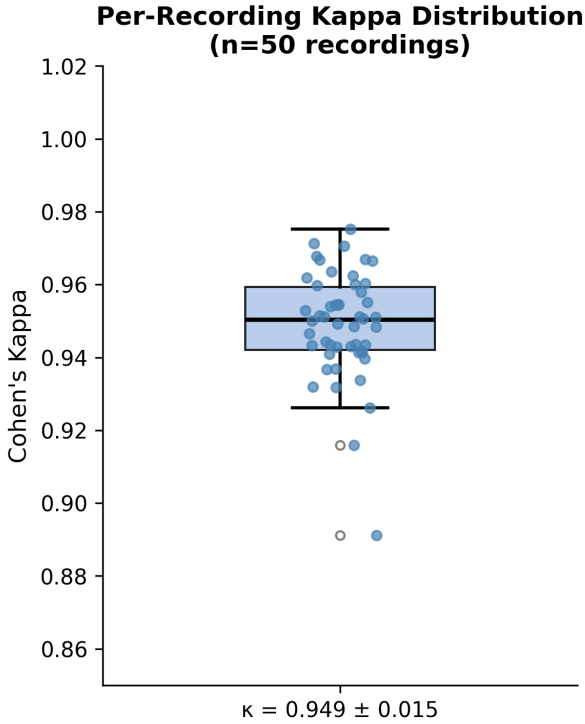

Figure 2: Distribution of Cohen's kappa values across all 50 recordings. Box shows median and interquartile range; individual recording values are overlaid.

Table 4: Aggregate validation metrics (mean ± SD across 50 recordings). Calibration epochs (360 per recording) were excluded from all comparisons.

Metric	Mean	SD
Cohen's kappa	0.9490	0.0148
Accuracy	97.25%	0.72%
Wake precision	95.90%	—
Wake recall	96.60%	—
Wake F1	0.9623	0.0174
NREM precision	98.12%	—
NREM recall	97.17%	—
NREM F1	0.9763	0.0050
REM precision	95.54%	—
REM recall	99.64%	—
REM F1	0.9754	0.0093

The mean accuracy of 97.25% is consistent with the benchmark of approximately 96.8% reported in the original AccuSleePy publication for the same dataset and pre-trained model [Barger et al., 2019], confirming that the pipeline is executing as expected.

3.2 Quality Control Summary

Zero of 50 recordings were flagged by any quality control criterion. No recording exceeded the Wake > 80% or REM > 25% thresholds, and no recording had NREM below 10%. No unbroken stage runs exceeding 60 minutes were detected. The maximum observed Wake proportion across all recordings was 53.5%, the maximum REM was 15.3%, and the minimum NREM was 38.2%, all of which fall comfortably within the physiologically plausible bounds.

A total of 481 low-confidence epochs (confidence score ≤ 0.8) were identified across all 50 recordings, representing a mean of 0.167% of total epochs per recording. The per-recording low-confidence count had a median of 8 epochs (range: 1–28). Per-recording low-confidence epoch lists are available in the `low_confidence_epochs/` directory for manual review if desired. No recordings were excluded from the analysis.

3.3 Sleep Architecture

The following results characterize the group-level sleep architecture of the 10 animals across all 50 recordings. These metrics constitute the primary analytical deliverable of the AccuSleePy analysis. All metrics are computed from the AccuSleePy predicted labels.

3.3.1 Stage Proportions

Figure 3 shows the group-level percentage of recording time spent in each sleep stage. Each data point represents one animal's mean across its 5 recordings. The three stages accounted for $34.53 \pm 5.16\%$ (Wake), $54.64 \pm 4.00\%$ (NREM), and $10.83 \pm 1.37\%$ (REM) of total recording time (mean ± SD across 10 animals).

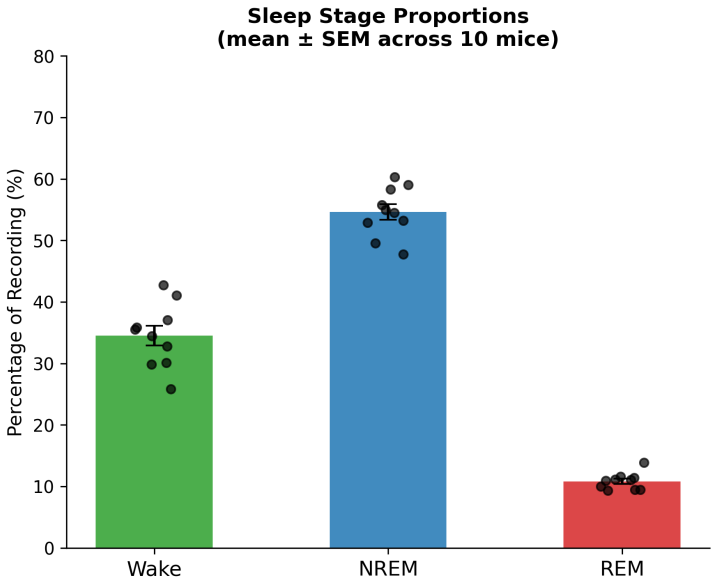

Figure 3: Group-level stage proportions. Bars show mean ± SEM across 10 animals. Individual animal means (averaged over 5 recordings per mouse) are overlaid as points. Wake = green, NREM = blue, REM = red.

3.3.2 Bout Duration Analysis

Figure 4 shows the distribution of mean bout duration per stage across animals. Mean bout durations were 41.5 ± 10.4 s (Wake), 62.9 ± 7.0 s (NREM), and 76.4 ± 11.1 s (REM; mean $\pm$ SD across 10 animals).

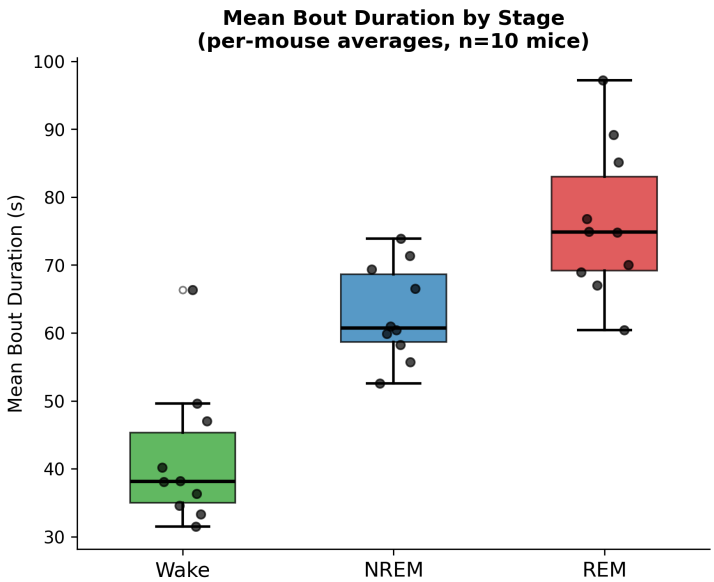

Figure 4: Mean bout duration (seconds) per stage across animals. Box plots show median and interquartile range; individual animal means are overlaid. Wake = green, NREM = blue, REM = red.

3.3.3 State Transition Probabilities

Figure 5 shows the mean state transition probability matrix across all 50 recordings. Diagonal values (self-transitions, i.e., the probability of remaining in the same stage at the next epoch boundary) are large, reflecting the temporal stability of sleep stages. Off-diagonal entries reveal the relative likelihood of transitions between stages.

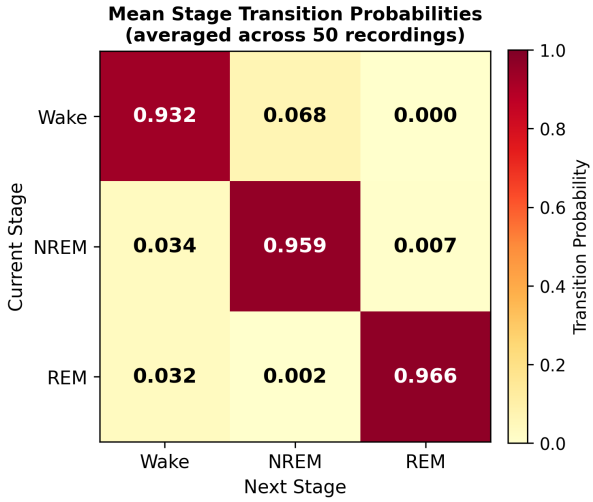

Figure 5: Mean state transition probability matrix (averaged across all 50 recordings). Entry (i, j) gives the probability that a transition out of stage i leads to stage j. Rows are normalized to sum to 1.

3.3.4 Representative Hypnograms

Figures 6–11 show the predicted sleep stage sequence (hypnogram) for the Day 1 recording of each of the six representative animals (Mouse01–Mouse06). The x-axis represents time in hours; the y-axis represents predicted stage. These hypnograms illustrate the typical alternating pattern of Wake, NREM, and REM sleep characteristic of rodent daytime recordings.

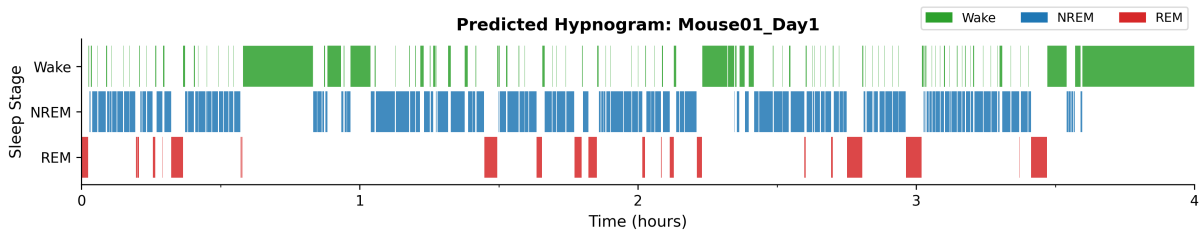

Figure 6: Predicted hypnogram: Mouse01, Day 1 (4-hour recording).

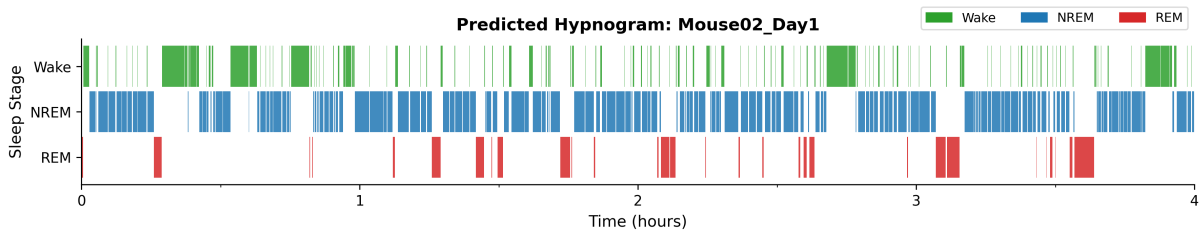

Figure 7: Predicted hypnogram: Mouse02, Day 1.

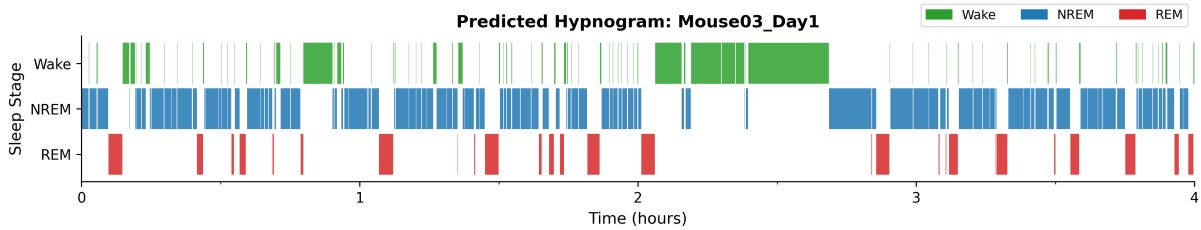

Figure 8: Predicted hypnogram: Mouse03, Day 1.

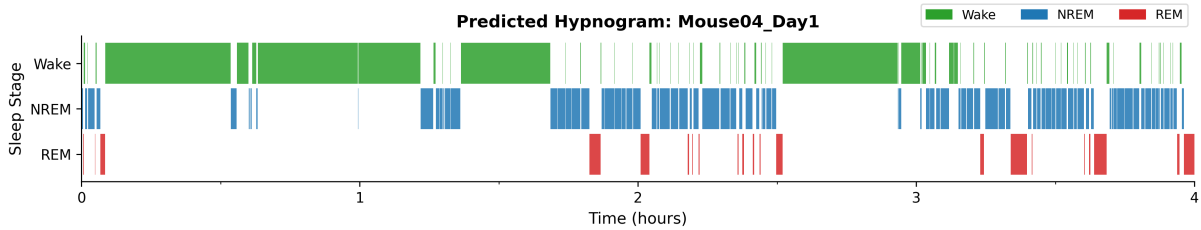

Figure 9: Predicted hypnogram: Mouse04, Day 1.

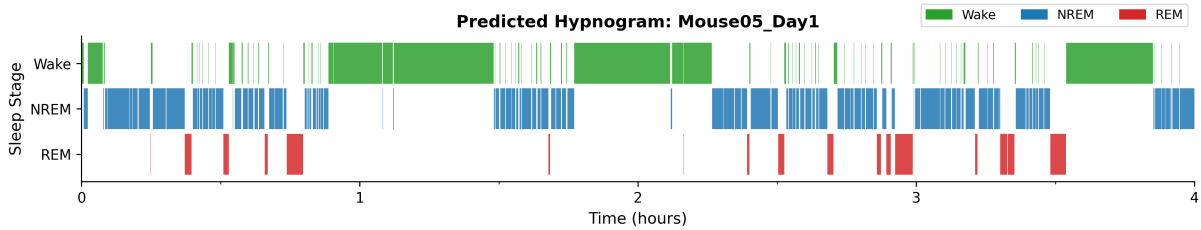

Figure 10: Predicted hypnogram: Mouse05, Day 1.

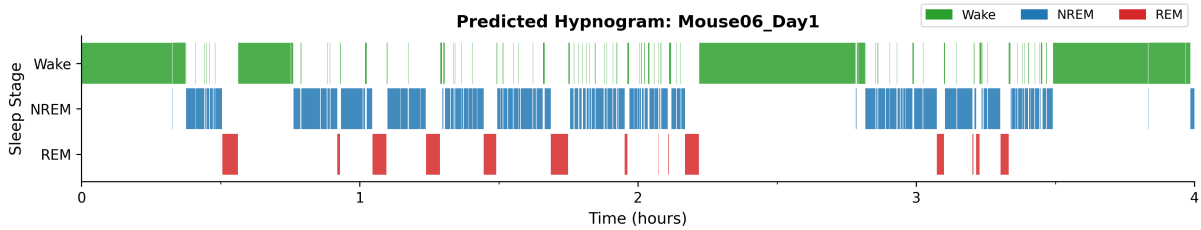

Figure 11: Predicted hypnogram: Mouse06, Day 1.

4 Limitations

- This analysis uses the standard single-channel cortical EEG and nuchal EMG configuration for which AccuSleePy was designed and validated. Model performance may differ for non-standard electrode configurations or multi-region recording setups.
- All analyses are descriptive. Stage proportions, bout structures, and transition probabilities characterize what the data show. Biological interpretation of these results is the responsibility of the researcher; no mechanistic claims are made here.

- No recordings were excluded from the analysis. Flagged recordings (none in this dataset) are noted in `QC_report.md` and Section 3.2 but remain included in all reported metrics. The decision to exclude any recording based on QC findings is left to the researcher.
- The validation comparison to expert manual labels is included as a technical quality check confirming correct pipeline execution. It is not intended to be interpreted as a primary scientific result of this analysis.

5 References

Barger, Z., Frye, C. G., Liu, D., Dan, Y., & Bouchard, K. E. (2019). Robust, automated sleep scoring by a compact neural network with distributional shift correction. *PLOS ONE*, 14(12), 1–18. https://doi.org/10.1371/journal.pone.0224642

Barger, Z., & Frye, C. (2025, June 6). AccuSleep. https://doi.org/10.17605/OSF.IO/PY5EB